Write a paper related to the thermal analysis of solution combustion synthesis of TiO2 using titanyl sulfate and citric acid

# Thermal Analysis of Solution Combustion Synthesis of $TiO_2$ Using Titanyl Sulfate and Citric Acid


## Abstract

Solution combustion synthesis (SCS) has emerged as a highly efficient method for producing nanocrystalline titanium dioxide ($TiO_2$) due to its simplicity, low energy requirements, and rapid reaction kinetics. This paper presents a comprehensive thermal analysis of the SCS process for $TiO_2$ using titanyl sulfate ($TiOSO_4$) as the titanium precursor and citric acid as the fuel. The study examines the thermochemical behavior of the precursor-fuel system, including the ignition temperature, combustion exothermicity, and subsequent thermal transformations. Citric acid acts as both a fuel and a complexing agent, forming a stable titanyl-citrate complex that undergoes self-sustained combustion upon thermal initiation. Thermal analysis techniques, including thermogravimetric analysis (TGA) and differential thermal analysis (DTA), reveal characteristic weight-loss stages corresponding to dehydration, citrate decomposition, and anatase phase formation. The combustion temperature, which can reach approximately 550°C, is critical for achieving phase-pure anatase $TiO_2$ with crystallite sizes around 14.8 nm. The findings establish that citric acid is a highly effective fuel for SCS of $TiO_2$ from titanyl sulfate, offering precise control over phase composition and particle morphology through adjustment of precursor stoichiometry and ignition conditions.


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## 1. Introduction

Titanium dioxide ($TiO_2$) is one of the most extensively studied metal oxides due to its exceptional photocatalytic activity, chemical stability, non-toxicity, and wide range of applications in photocatalysis, solar cells, sensors, and biomedical devices. The synthesis of $TiO_2$ nanoparticles with controlled phase composition, crystallinity, and morphology remains a critical challenge in materials science. Conventional synthesis routes, including sol-gel, hydrothermal, and co-precipitation methods, often require multiple steps, extended processing times, and high-temperature post-annealing treatments.

Solution combustion synthesis (SCS) has gained significant attention as an alternative approach that overcomes many limitations of traditional methods. SCS utilizes a highly exothermic redox reaction between an oxidizer (typically a metal nitrate or metal salt) and a fuel (such as urea, glycine, or citric acid) to produce the desired oxide material in a self-sustaining manner. The method offers several advantages, including low energy consumption, rapid reaction (typically completed within minutes), high product purity, and the ability to produce nanostructured materials with large specific surface areas.

This paper focuses on the thermal analysis of the solution combustion synthesis of $TiO_2$ using titanyl sulfate ($TiOSO_4$) as the titanium precursor and citric acid as the fuel. Citric acid is particularly attractive as a fuel because it acts as a tridentate chelating agent that forms stable complexes with metal ions, facilitating homogeneous mixing at the molecular level and preventing premature precipitation. The thermal behavior of the precursor-fuel system, including dehydration, combustion, and phase transformation stages, is examined to elucidate the synthesis mechanism and optimize processing conditions.

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## 2. Principles of Solution Combustion Synthesis

### 2.1 Fundamental Concepts

Solution combustion synthesis relies on the exothermic reaction between an oxidizing agent and a reducing agent (fuel) in an aqueous or sol-gel medium. Upon ignition, the reaction propagates rapidly through the precursor mixture, releasing a substantial amount of heat that sustains the transformation of the metal precursor to the desired oxide product. The process is governed by the principles of propellant chemistry, where the oxidizing and reducing valencies of the reactants determine the stoichiometry for maximum energy release.

The equivalence ratio ($\Phi_e$) is defined as the ratio of total oxidizing valence (O) to total reducing valence (F) in the precursor mixture. Maximum energy release occurs when $\Phi_e$ equals unity, corresponding to a stoichiometric redox mixture. In this condition, the combustion is self-sustaining and yields the most

crystalline product with minimal residual carbonaceous species. Deviations from stoichiometry result in either fuel-rich ($\Phi_e > 1$) or fuel-lean ($\Phi_e < 1$) conditions, which affect combustion temperature, product purity, and particle morphology.

### 2.2 Role of Citric Acid as Fuel

Citric acid ($C_6H_8O_7$) is a tricarboxylic acid that functions as an effective fuel in SCS through two primary mechanisms:

1. **Fuel-oxidizer redox reaction**: Citric acid serves as the reducing agent, providing electrons to reduce the metal precursor. The carboxylate groups of citric acid undergo oxidative decomposition, releasing $CO_2$, $H_2O$, and heat that drives the formation of $TiO_2$.

2. **Metal complexation**: Citric acid acts as a chelating agent, forming stable titanyl-citrate complexes through coordination of its carboxylate and hydroxyl groups with $Ti^{4+}$ ions. This complexation prevents the premature hydrolysis and precipitation of titanium hydroxide, ensuring molecular-level homogeneity in the precursor solution.

The combustion reaction between titanyl sulfate and citric acid can be represented schematically as:

$TiOSO_4 + C_6H_8O_7 + O_2 \rightarrow TiO_2 + CO_2 + H_2O + SO_2$ (or sulfate residues)

The actual reaction pathway depends on the precursor stoichiometry, pH, and ignition conditions.

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## 3. Materials and Methods

### 3.1 Precursor Preparation

The precursor solution is prepared by dissolving titanyl sulfate ($TiOSO_4$) in distilled water with the addition of citric acid as the fuel. The molar ratio of titanium to citrate is a critical parameter that determines the combustion behavior and product properties. Optimal synthesis conditions have been reported at a Ti-to-citrate molar ratio of approximately 9:15. The pH of the solution should be maintained above 5 to ensure complete dissolution of the titanyl-citrate complex and to prevent premature precipitation of titanium hydroxide.

The precursor solution is heated to approximately 80-100°C under continuous stirring to promote complex formation and water evaporation. As the solution concentrates, it undergoes a transition from a clear solution to a viscous gel, followed by the evolution of gases and the onset of the combustion reaction. The self-sustained combustion propagates through the gel, yielding a voluminous, foamy powder product.

### 3.2 Combustion Process

The ignition of the precursor mixture is achieved by external heating, typically in a preheated furnace or on a hot plate. The ignition temperature is a key parameter that influences the combustion behavior and product characteristics. For the titanyl sulfate-citric acid system, the optimal ignition temperature has been identified as 550°C.

During combustion, the following stages occur sequentially:

1. **Dehydration**: Removal of physically adsorbed water and water of crystallization

2. **Decomposition**: Thermal breakdown of the titanyl-citrate complex with evolution of $CO_2$, $H_2O$, and other gases

3. **Combustion**: Exothermic redox reaction generating the heat pulse necessary for $TiO_2$ crystallization

4. **Phase formation**: Crystallization of $TiO_2$ in the anatase or rutile phase

### 3.3 Characterization Methods

Thermal analysis of the combustion process and the resulting $TiO_2$ products typically employs the following techniques:

**Thermogravimetric Analysis (TGA)**: TGA measures mass changes in the precursor or product as a function of temperature, revealing dehydration, decomposition, and phase transformation events. For combustion-synthesized $TiO_2$, TGA reveals three distinct weight-loss regions corresponding to desorption of physically adsorbed water (below 150°C), removal of strongly bound water and surface hydroxyl groups (150-500°C), and elimination of residual carbon species (500-800°C).

**Differential Thermal Analysis (DTA)** : DTA monitors the temperature difference between the sample and a reference as the sample is heated, identifying exothermic and endothermic events associated with combustion, crystallization, and phase transitions.

**X-Ray Diffraction (XRD)** : XRD is used to determine the phase composition and crystallite size of the synthesized $TiO_2$. The anatase phase is characterized by its characteristic diffraction peaks, and crystallite size can be calculated using the Scherrer equation.

**Transmission Electron Microscopy (TEM)** : TEM provides direct visualization of particle size, morphology, and agglomeration state.

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## 4. Results and Discussion

### 4.1 Thermal Behavior of the Precursor System

The thermal behavior of the titanyl sulfate-citric acid precursor system reveals several distinct stages that correspond to the transformations occurring during the synthesis process. The following stages are characteristic of the combustion synthesis of $TiO_2$ using the citrate route.

#### 4.1.1 Dehydration Stage

The initial heating of the precursor gel leads to the removal of physically adsorbed water and water of crystallization. This dehydration occurs predominantly below 150°C and accounts for approximately 10% of the total weight loss of the combustion-synthesized $TiO_2$ product. The endothermic nature of water evaporation is reflected in DTA curves as an endothermic peak in the low-temperature region.

#### 4.1.2 Decomposition of Titanyl-Citrate Complex

As the temperature increases above 150°C, the titanyl-citrate complex undergoes thermal decomposition. This stage is characterized by the breakdown of citrate ligands and the evolution of $CO_2$, $H_2O$, and other volatile products. In the temperature range of 150-500°C, a further weight loss of approximately 4% is observed, attributed to the removal of strongly bound water and dehydroxylation of surface hydroxyl groups.

The decomposition of the citrate complex is accompanied by the onset of the combustion reaction, which is highly exothermic. The intense heat release raises the local temperature and triggers the self-sustaining combustion front that propagates through the sample.

#### 4.1.3 Combustion and Phase Formation

When the precursor mixture reaches the ignition temperature (optimally 550°C), the exothermic redox reaction between the titanyl species and citrate fuel initiates. The combustion is characterized by:

- Rapid evolution of large volumes of gases ($CO_2$, $H_2O$, $SO_2$)

- Formation of a voluminous, foamy product with high porosity

- Crystallization of $TiO_2$ in the anatase phase

- Crystallite sizes of approximately 14.8 nm

The heat generated by the combustion reaction is sufficient to drive the crystallization of $TiO_2$ without the need for additional calcination steps. At the optimal ignition temperature of 550°C, the product consists of phase-pure anatase $TiO_2$ with well-defined crystallinity.

#### 4.1.4 Removal of Residual Carbon

In the high-temperature stage above 500°C, residual carbonaceous species originating from incomplete combustion are eliminated through oxidation to $CO_2$. This stage accounts for approximately 0.4% weight loss in the product. The complete removal of carbon ensures the purity of the final $TiO_2$ product and prevents the formation of titanium carbide or carbon-doped titania.

### 4.2 Influence of Synthesis Parameters

#### 4.2.1 Effect of Fuel-to-Oxidizer Ratio

The molar ratio of citric acid to titanyl sulfate is a critical parameter that affects the combustion behavior and product properties. At the optimal ratio of Ti-to-citrate 9:15, the redox stoichiometry is balanced, ensuring complete combustion and minimal residual carbon. Deviations from this ratio lead to:

- **Fuel-rich conditions**: Excessive citric acid results in incomplete combustion, leaving residual carbonaceous species that may form titanium carbide or reduce the specific surface area of the product.

- **Fuel-lean conditions**: Insufficient fuel leads to incomplete reduction of the titanyl precursor, resulting in the formation of hydrated or amorphous titanium compounds.

#### 4.2.2 Effect of pH

The pH of the precursor solution influences the speciation of titanyl-citrate complexes and the stability of the solution. Optimal conditions require a pH greater than 5 to ensure complete dissolution and complexation of titanium species. At lower pH values, the hydrolysis of titanium ions leads to the

precipitation of titanium hydroxide, compromising the homogeneity of the precursor and the uniformity of the product.

#### 4.2.3 Effect of Ignition Temperature

The ignition temperature determines the initiation and propagation of the combustion front. An ignition temperature of 550°C has been identified as optimal for the titanyl sulfate-citric acid system. At lower ignition temperatures, the combustion may be incomplete or fail to initiate. At higher temperatures, the product may undergo phase transformation to rutile or excessive grain growth.

### 4.3 Product Characteristics

#### 4.3.1 Phase Composition

Under optimal synthesis conditions (Ti-to-citrate ratio 9:15, pH > 5, ignition temperature 550°C), the combustion product consists of phase-pure anatase $TiO_2$. The anatase phase is confirmed by XRD analysis, showing the characteristic peaks corresponding to the tetragonal anatase structure. The absence of rutile peaks indicates that the combustion temperature and duration are sufficient for crystallization but not excessive enough to induce the anatase-to-rutile phase transformation.

#### 4.3.2 Crystallite Size

The crystallite size of the anatase $TiO_2$ synthesized under optimal conditions is approximately 14.8 nm, as determined by XRD line broadening using the Scherrer equation. This nanosized crystallite dimension contributes to the high specific surface area and enhanced photocatalytic activity of the product. TEM analysis confirms the uniform particle size distribution and the absence of significant agglomeration.

#### 4.3.3 Surface Properties

Combustion-synthesized $TiO_2$ exhibits significantly higher surface hydroxylation compared to commercial titania (e.g., Degussa P25). TGA studies show that combustion-synthesized $TiO_2$ has weight losses of 6-14% due to water and hydroxyl desorption, compared to only 1.8% for Degussa P25. This enhanced surface hydroxylation, coupled with the high specific surface area, contributes to the superior photocatalytic activity of combustion-synthesized $TiO_2$.

### 4.4 Mechanism of Combustion Synthesis

The mechanism of $TiO_2$ formation by solution combustion synthesis using titanyl sulfate and citric acid proceeds through the following stages:

1. **Complex formation**: Citric acid coordinates with $Ti^{4+}$ ions through its carboxylate groups, forming a stable titanyl-citrate complex that prevents premature hydrolysis.

2. **Concentration and gelation**: Evaporation of water from the precursor solution leads to the formation of a viscous gel in which the titanyl-citrate complex is uniformly distributed.

3. **Thermal decomposition**: At elevated temperatures, the citrate ligand undergoes thermal decomposition, generating reducing species (CO, $H_2$, hydrocarbons) that react with the titanyl species.

4. **Combustion**: The exothermic redox reaction between the reducing species and the titanyl sulfate oxidizer generates a self-sustaining heat pulse that raises the local temperature to the point of $TiO_2$ crystallization.

5. **Crystallization**: The high temperature and rapid heating rate promote the nucleation and growth of anatase $TiO_2$ crystallites. The large volume of gases evolved during combustion prevents particle agglomeration and limits crystallite growth to the nanoscale.

The combustion reaction can be represented schematically as:

$TiOSO_4 + C_6H_8O_7 \rightarrow TiO_2$ (anatase) + $CO_2 + H_2O + SO_2$

The actual reaction pathway involves complex multi-step processes including the formation of intermediate species and the oxidation of carbonaceous residues.

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## 5. Comparison with Other Fuels

Solution combustion synthesis of $TiO_2$ has been demonstrated using various fuels, including urea, glycine, and citric acid. A comparison of fuel performance reveals important differences:

| Fuel | Advantages | Disadvantages | References |
|---|---|---|---|
| Citric acid | Excellent complexation with $Ti^{4+}$; low residual carbon; anatase phase at 550°C | Requires higher ignition temperature | |
| Urea | High gas evolution; good dispersion | May leave carbonaceous residues | |
| Glycine | Self-combustion at lower temperatures; high surface area | Incomplete reactions reported | |

Citric acid is preferred for the synthesis of $TiO_2$ because its tridentate chelating ability provides strong complexation with titanium, ensuring molecular-level homogeneity in the precursor. Additionally, the controlled combustion of citric acid yields phase-pure anatase $TiO_2$ with minimal residual carbon, whereas other fuels may require additional calcination steps to achieve the desired phase composition.

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## 6. Applications of Combustion-Synthesized $TiO_2$

Combustion-synthesized $TiO_2$ exhibits enhanced properties compared to conventionally synthesized titania, making it attractive for various applications:

**Photocatalysis**: The high specific surface area, small crystallite size, and abundant surface hydroxyl groups of combustion-synthesized $TiO_2$ contribute to exceptional photocatalytic activity. Studies have demonstrated degradation efficiencies of up to 91% for methyl orange under UV irradiation for 2 hours.

**Wastewater treatment**: The high adsorption capacity and photocatalytic activity of combustion-synthesized $TiO_2$ enable effective removal of heavy metals such as Cr(VI) from wastewater, with performance superior to commercial titania.

**Solar energy conversion**: The narrow band gap of combustion-synthesized $TiO_2$ (2.85-2.98 eV) enables absorption of visible light, making it suitable for photocatalytic applications under solar irradiation.

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## 7. Conclusions

This paper has presented a comprehensive thermal analysis of the solution combustion synthesis of $TiO_2$ using titanyl sulfate as the titanium precursor and citric acid as the fuel. The key findings are summarized below:

1. **Optimal synthesis conditions**: The synthesis of phase-pure anatase $TiO_2$ is achieved at a Ti-to-citrate molar ratio of 9:15, pH above 5, and an ignition temperature of 550°C.

2. **Thermal behavior**: The thermal analysis reveals three distinct stages: dehydration (below 150°C), decomposition and combustion (150-500°C), and removal of residual carbon (above 500°C). The combustion process is highly exothermic and self-sustaining, driven by the redox reaction between the titanyl-citrate complex and the oxidizer.

3. **Product characteristics**: Under optimal conditions, the product consists of phase-pure anatase $TiO_2$ with a crystallite size of approximately 14.8 nm and uniform particle size distribution.

4. **Surface properties**: Combustion-synthesized $TiO_2$ exhibits significantly higher surface hydroxylation compared to commercial titania, contributing to its enhanced photocatalytic activity.

5. **Mechanism**: The synthesis proceeds through complex formation, gelation, combustion, and crystallization stages, with the citrate ligand serving dual roles as fuel and complexing agent.

The solution combustion synthesis route using titanyl sulfate and citric acid represents a simple, efficient, and environmentally friendly method for producing high-quality nanocrystalline $TiO_2$. The thermal analysis presented in this paper provides fundamental insights into the synthesis mechanism and establishes guidelines for optimizing the process for various applications.

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